MD-DGFLS: MULTI-DOMAIN DEEPFAKE FORENSIC VERIFICATION CERTIFICATE

IEEE TPAMI / CVPR Technical Forensics Specification v2.2-PROD | Cryptographic Audit Dossier

File Name:	49dd4745.webp	**Target Domain:**	Face
Audit Timestamp:	2026-09-28 05:29:42 UTC	**SHA-256 Hash:**	3a7321ad8fee702cba4f17ca...
Forensic Verdict:	**VERIFIED AUTHENTIC MEDIA**	**Confidence:**	**56.9%**
Tamper Type:	Authentic Media Stream	**Bounding Box:**	None

Three-Stage Visual Forensics Pipeline



Stage 1: Source Image



Stage 2: Tamper Overlay



Stage 3: XAI Heatmap

Mathematical Innovations & Multi-Expert Telemetry

CDFAM (Cross-Domain Alignment)	Contrastive Score: 0.481 (Margin: m=0.3)
ARCC (Entropy Calibration)	Dynamic Calibration Temperature: T = 1.031
TGHFL (Transformer-GAN Cross-Attention)	Cross-Attention Energy: 0.0005
MEFCN (Multi-Expert Consensus)	Face: 78.9% | Medical: 5.1% | Satellite: 6.6% | Regular: 4.7% | Natural: 4.7%
HLRN (Localization Refinement)	Coarse-to-Fine Refinement: 1280x720 Native
Frequency Stream (ELA & Laplacian)	ELA Discrepancy: 0.727 | Sensor Noise Var: 70.7

This certificate is cryptographically anchored and generated by the MD-DGFLS Autonomous Forensics Engine. Verified under IEEE TPAMI standards.